# Zixiang Xu


## Contact Information

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**Ph.D. Student**
Computer Science
University of Southern California
Los Angeles, CA


## Education

**University of Southern California (USC)** — *Los Angeles, CA*
*Ph.D. Student in Computer Science* · Advisor: Prof. Yue Zhao — Aug. 2026 – Present
FORTIS Lab. Expected graduation: 2030.

**Sichuan University (SCU)** — *Chengdu, China*
*B.S. in Computer Science* — 2022 – 2026

## Research Experience

**Meta AI** — *Remote*
*Research Collaboration* — Sep. 2026 – Present
Lead Grid-Bench, building executable spatial environments for frontier-model evaluation and post-training.

**Tencent (WeChat AI)** — *Beijing, China*
*Research Intern* · Foundation Model Group — Jun. 2026 – Aug. 2026
Led AlgoWorlds: 240 verifiable tool-use worlds across ten optimization families; evaluated seven LLMs and generated agent trajectories for downstream post-training.

**Alibaba Group** — *Beijing, China*
*Research Intern* — Mar. 2026 – May 2026
Investigated representation-level mechanisms behind LLM-as-a-judge scoring bias and led a mechanistic interpretability study.

**MBZUAI** — *Abu Dhabi, UAE*
*Research Intern* · Supervisor: Prof. Xiuying Chen — Mar. 2025 – Aug. 2025
Led research on trustworthy LLMs and AI agents, including multilingual robustness, social reasoning, and dynamic agentic workflows; contributed to ACL, NeurIPS, and EMNLP papers.

**MINE Lab, University of Notre Dame** — *Remote*
*Research Intern* · Supervisor: Prof. Xiangliang Zhang — Oct. 2024 – Aug. 2025
Co-led research on robust reasoning, contextual reliability, and multilingual evaluation, resulting in publications at ACL and NeurIPS.

**MIT Center for Transportation & Logistics** — *Massachusetts, USA*
*Summer Research Student* · Supervisor: Prof. Elenna R. Dugundji — Aug. 2024
Led Temporal-IRL on port congestion and berth scheduling; selected for a TRB 2025 oral presentation.

**Microsoft (Asia) Internet Engineering Institute** — *Beijing, China*
*Research Intern* · Supervisor: Wang Pan — Jan. 2024 – Mar. 2024
Evaluated and analyzed LLMs and vision-language models, including GPT-4V assessment and capability reproduction.

## Honors and Awards

**USC Annenberg Fellowship** — 2026
A competitive fellowship top-off awarded to selected Ph.D. students in the USC Viterbi School of Engineering.

**CCF Elite Collegiate Award** — 2024
One of 100 junior and senior undergraduates selected nationwide; recognized for academics, research, and service. Invited speaker/session host at CNCC; team placed **3rd nationwide** in the CCSP Team Contest.

**Gold Medal, The 2022 ICPC Asia Jinan Regional Contest** — 2022
Gold medal in the ICPC, the world's premier collegiate algorithmic programming competition.

**Silver Medal, National Olympiad in Informatics (NOI)** — 2020
Silver medal in China's national informatics olympiad for secondary-school students, organized by the China Computer Federation (CCF).

## Publications

**[1] Grid-Bench: Can LLMs Predict How Discrete Worlds Change?**
**Zixiang Xu**, Zirui Song, Wanrong Zheng, Chiyu Zhang, Yue Huang, Yuyang Peng, Kexin Zhang, Jinkun Chen, Xiuying Chen, Xiangliang Zhang, Yan Liu, Laurent Itti, Frank Shyu, Xiaoqiang Luo, Sandeep Pandey, Xi Liu, Luke Simon, Xiyang Hu, Yue Zhao.
*Preprint, 2026.* [Project] [Paper] [Code] [Data]

**[2] JevOut: Natural Context Can Flip Decision Models**
**Zixiang Xu**, Zirui Song, Chiyu Zhang, Xiuying Chen, Xi Liu, Xiyang Hu, Yue Zhao.
*Preprint, 2026.* **Agenthon @ NeurIPS 2026** (Poster).
[Project] [Paper] [Code & Data]

**[3] SocialMaze: A Benchmark for Evaluating and Enhancing Social Reasoning in Large Language Models in Complex Social Environments**
**Zixiang Xu**, Yanbo Wang, Yue Huang, Haomin Zhuang, Yujun Zhou, Jiayi Ye, et al.
**Findings of EMNLP 2026**; previously SocialSim @ **COLM 2025** Workshop (Spotlight).
[Project] [Paper] [Code] [Data]

**[4] Cross-Lingual Pitfalls: Automatic Probing Cross-Lingual Weakness of Multilingual Large Language Models**
**Zixiang Xu**[*], Yanbo Wang[*], Yue Huang[*], Xiuying Chen, Jieyu Zhao, Meng Jiang, Xiangliang Zhang ([*]Equal contribution).
**ACL 2025**. [Project] [Paper] [Code]

**[5] Adaptive Distraction: Probing LLM Contextual Robustness with Automated Tree Search**
Yanbo Wang[*], **Zixiang Xu**[*], Yue Huang[*], Chujie Gao, Siyuan Wu, Jiayi Ye, et al. ([*]Equal contribution).
**NeurIPS 2025**. [Paper] [Code]

**[6] DyFlow: Dynamic Workflow Framework for Agentic Reasoning**
Yanbo Wang, **Zixiang Xu**, Yue Huang, Xiangqi Wang, Zirui Song, Lang Gao, Chenxi Wang, et al.
**NeurIPS 2025**. [Paper] [Code]

**[7] AlgoWorlds: Benchmarking Tool Use for Global Optimization in Algorithmic Worlds**
**Zixiang Xu**, Jiaan Wang, Fandong Meng.
*Preprint, 2026.* [Project] [Paper] [Code & Data]

**[8] Inside the Unfair Judge: A Mechanistic Interpretability Account of LLM-as-Judge Bias**
**Zixiang Xu**, Sixian Li, Huaxing Liu, Xiang Wang, Shuai Li, Zirui Song, Xiuying Chen.
*Preprint, 2026.* [Project] [Paper]

**[9] GTA: Graph Theory Agent and Benchmark for Algorithmic Graph Reasoning with LLMs**
**Zixiang Xu**, Yanbo Wang, Chenxi Wang, Lang Gao, Zirui Song, Yue Huang, Zhaorun Chen, Xiangliang Zhang, Xiuying Chen.
*Preprint, 2026.* [Project] [Paper] [Code & Data]

## Publications (continued)

**[10] Do LLMs "Feel"? Emotion Circuits Discovery and Control**
Chenxi Wang, Yixuan Zhang, et al., **Zixiang Xu**, et al.
**ICML 2026**. [Paper] [Code]

**[11] Audio Jailbreak: An Open Comprehensive Benchmark for Jailbreaking Large Audio-Language Models**
Zirui Song, Qian Jiang, et al., **Zixiang Xu**, et al.
**ACL 2026**. [Paper] [Code]

**[12] ManipLVM-R1: Reinforcement Learning for Reasoning in Embodied Manipulation with Large Vision-Language Models**
Zirui Song, Guangxian Ouyang, et al., **Zixiang Xu**, et al.
**AAAI 2026**. [Paper]

**[13] Temporal-IRL: Modeling Port Congestion and Berth Scheduling with Inverse Reinforcement Learning**
Yikuan Hu*, **Zixiang Xu***, Wei Zhang*, et al. (*Equal contribution).
**TRB 2025** (Oral). [Paper]

**[14] Under the Shadow of Babel: How Language Shapes Reasoning in LLMs**
Chenxi Wang, Yixuan Zhang, Lang Gao, **Zixiang Xu**, et al.
**Findings of EMNLP 2025**. [Paper]

**[15] GenUDC: High-Quality 3D Mesh Generation with Unsigned Dual Contouring Representation**
Ruowei Wang, Jiaqi Li, et al., **Zixiang Xu**, et al.
**ACM MM 2024**. [Paper]

**[16] Beyond Survival: Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies**
Zirui Song, Yuan Huang, Junchang Liu, et al., **Zixiang Xu**, et al.
*Preprint, 2025.* [Paper]

**[17] Evaluate Bias without Manual Test Sets: A Concept Representation Perspective for LLMs**
Lang Gao, Kaiyang Wan, Wei Liu, et al., **Zixiang Xu**, et al.
*Preprint, 2025.* [Paper]

**[18] MTFusion: Reconstructing Any 3D Object from Single Image Using Multi-word Textual Inversion**
Yu Liu, Ruowei Wang, Jiaqi Li, **Zixiang Xu**, Qijun Zhao.
**PRCV 2024** (Oral). [Paper]

**[19] Text-Based Face Retrieval: Methods and Challenges**
Yuchuan Deng, Qijun Zhao, Zhanpeng Hu, **Zixiang Xu**.
**CCBR 2023** (Oral). [Paper]

## Reviewing Service

**Conference Reviewer:** AAAI 2027; COLM 2026; ICLR 2026; NeurIPS 2026; NLPCC 2025.

**ACL Rolling Review Reviewer:** February and May 2025 cycles.

**Workshop Reviewer:** GroundLM and REALM at EMNLP 2026; Social Simulation with LLMs at COLM 2025.